

WASIM RAJA

MUSIC
STORE

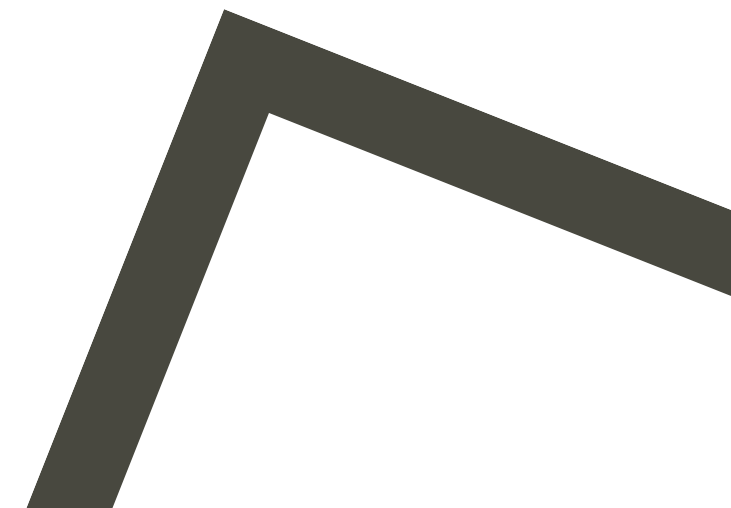
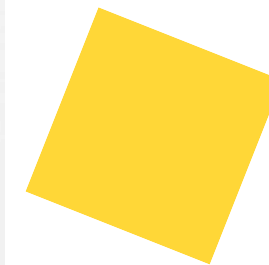


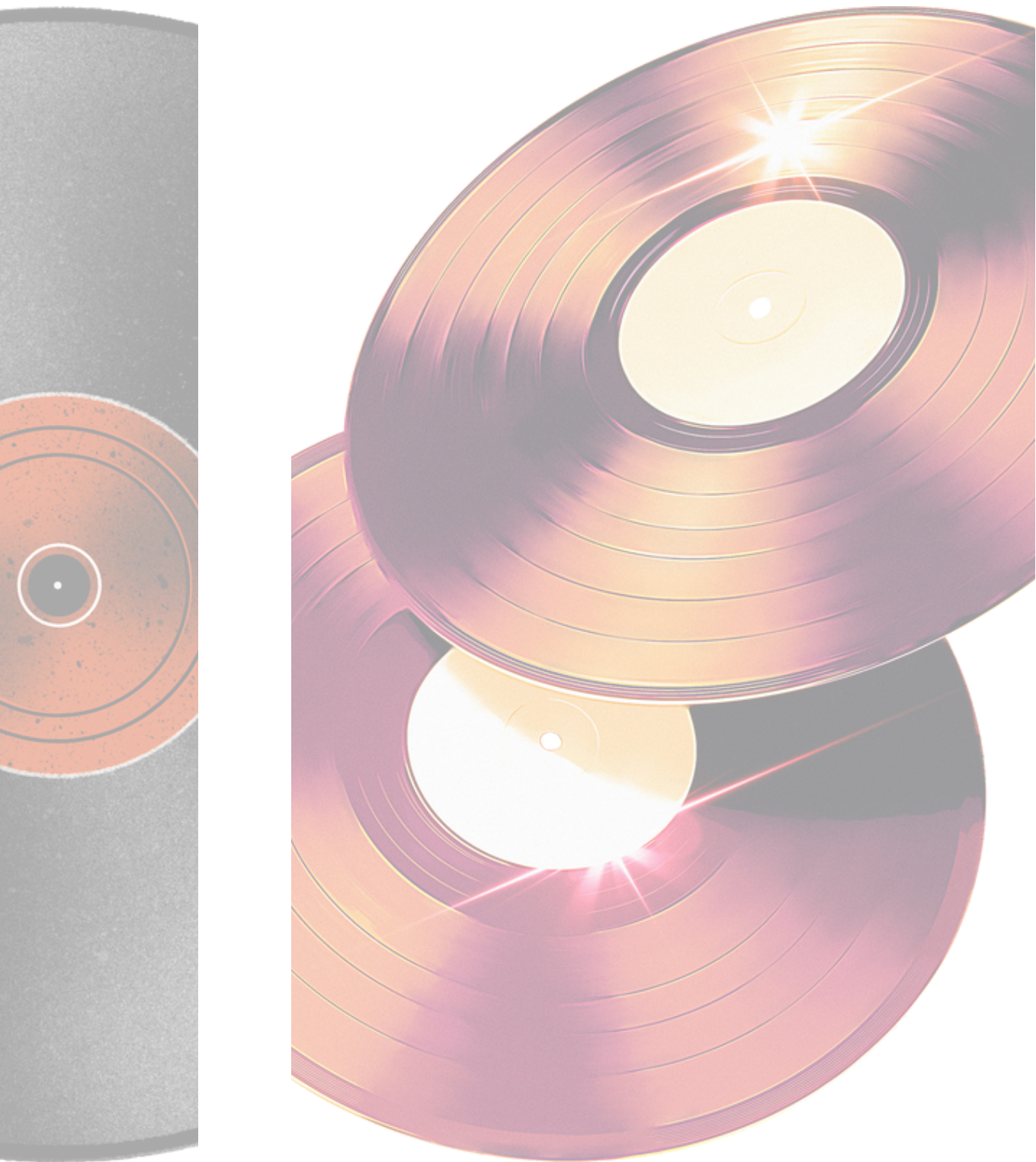
ANALYSIS

Transforming Data Into Insights

QUESTION SET - 1

EASY





**Q. 1. WHO IS THE SENIOR MOST
EMPLOYEE BASED ON JOB TITLE?**

```
SELECT  
FIRST_NAME,  
LAST_NAME,  
LEVELS  
FROM EMPLOYEE  
ORDER BY LEVELS DESC  
LIMIT 1
```

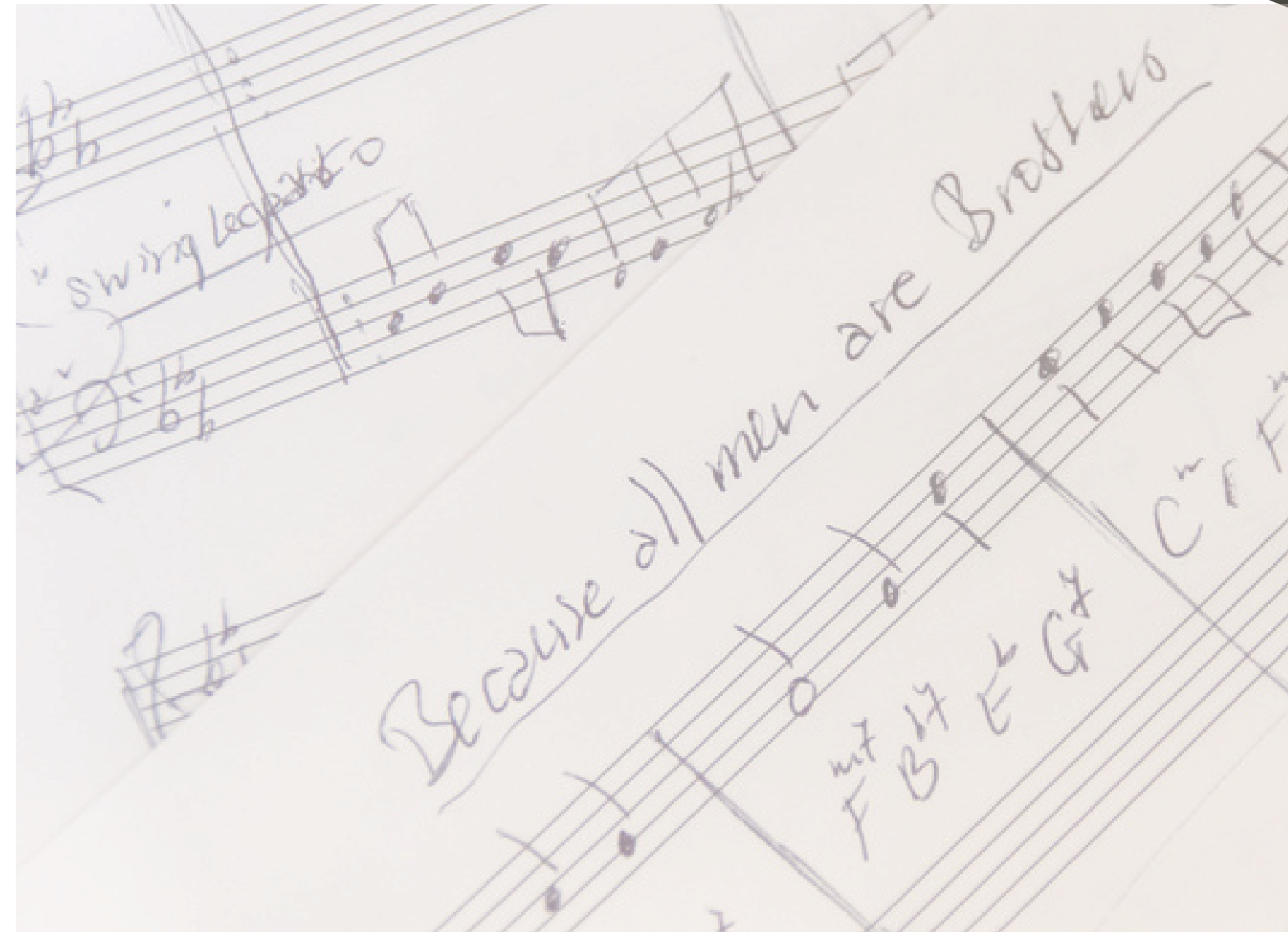
Q.2. WHICH COUNTRIES HAVE THE MOST INVOICES?

```
SELECT  
BILLING_COUNTRY,  
COUNT(INVOICE_ID) AS MOST_INV  
FROM INVOICE  
GROUP BY BILLING_COUNTRY  
ORDER BY MOST_INV DESC
```



Q.3. WHAT ARE TOP 3 VALUES OF TOTAL INVOICE?

**SELECT * FROM
INVOICE
ORDER BY TOTAL DESC
LIMIT 3**





Q.4. WHICH CITY HAS THE BEST CUSTOMERS? WE WOULD LIKE TO THROW A PROMOTIONAL MUSIC FESTIVAL IN THE CITY WE MADE THE MOST MONEY. WRITE A QUERY THAT RETURNS ONE CITY THAT HAS THE HIGHEST SUM OF INVOICE TOTALS. RETURN BOTH THE CITY NAME & SUM OF ALL INVOICE TOTALS.

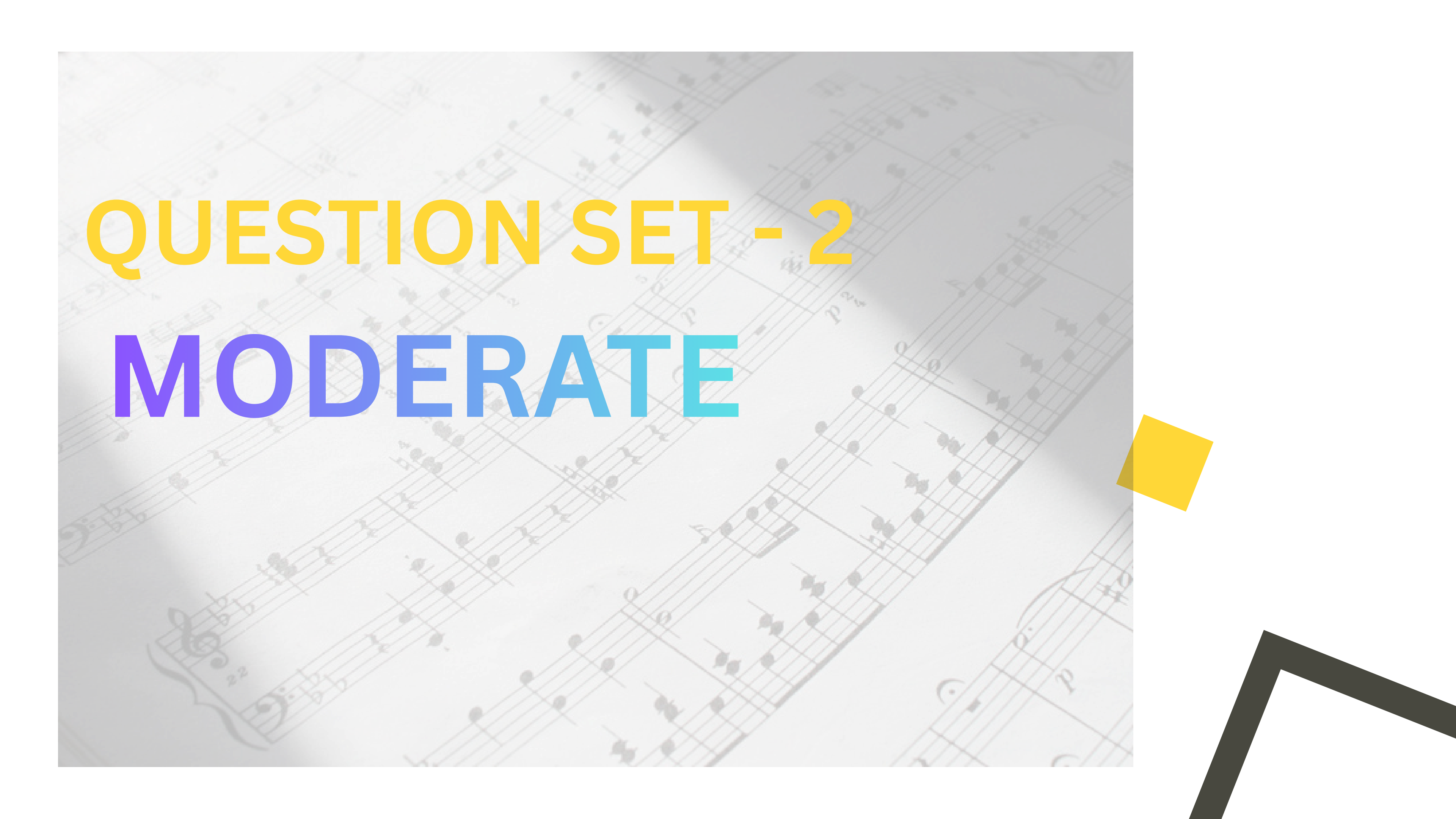
```
SELECT  
  BILLING_CITY, SUM(TOTAL) AS  
  HIGHEST_SUM  
FROM INVOICE  
GROUP BY BILLING_CITY  
ORDER BY HIGHEST_SUM DESC  
LIMIT 1
```



5. WHO IS THE BEST CUSTOMER? THE CUSTOMER WHO HAS SPENT THE MOST MONEY WILL BE DECLARED THE BEST CUSTOMER. WRITE A QUERY THAT RETURNS THE PERSON WHO HAS SPENT THE MOST MONEY.

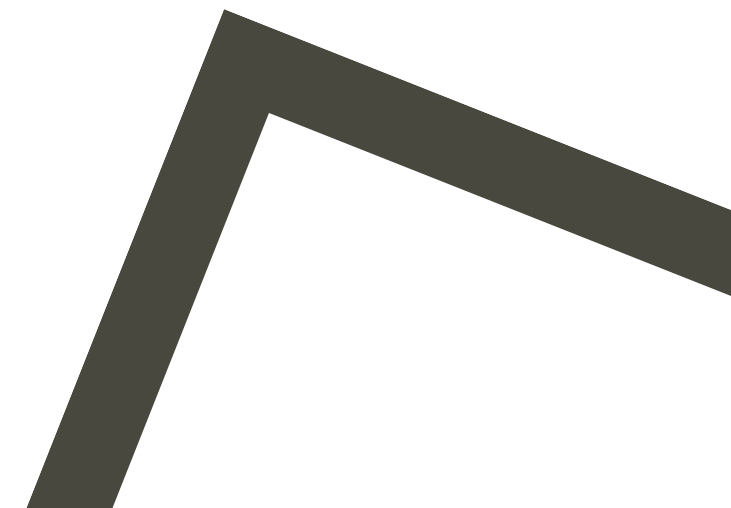
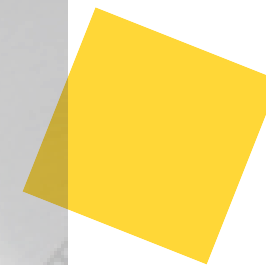
```
SELECT  
  C.CUSTOMER_ID, C.FIRST_NAME,  
  C.LAST_NAME,  
  INV.BILLING_CITY,  
  SUM(INV.TOTAL)AS HIGEST_SUM  
FROM CUSTOMER C  
LEFT JOIN INVOICE INV  
ON C.CUSTOMER_ID = INV.CUSTOMER_ID  
GROUP BY C.CUSTOMER_ID, INV.BILLING_CITY  
ORDER BY HIGEST_SUM DESC  
LIMIT 1
```



The background of the slide features a blurred, grayscale image of musical notation on staves, with various notes and clefs visible.

QUESTION SET - 2

MODERATE



**Q. 1. WRITE QUERY TO RETURN THE EMAIL,
FIRST NAME, LAST NAME, & GENRE OF ALL
ROCK MUSIC LISTENERS.
-- RETURN YOUR LIST ORDERED
ALPHABETICALLY BY EMAIL STARTING
WITH A.**

```
SELECT DISTINCT C.FIRST_NAME, C.LAST_NAME,  
C.EMAIL  
FROM CUSTOMER C  
JOIN INVOICE INV ON C.CUSTOMER_ID =  
INV.CUSTOMER_ID  
JOIN INVOICE_LINE INV_LINE ON INV.INVOICE_ID =  
INV_LINE.INVOICE_ID  
WHERE TRACK_ID IN(  
SELECT TRACK_ID FROM TRACK T  
JOIN GENRE G ON T.GENRE_ID = G.GENRE_ID  
WHERE G.NAME = 'ROCK' AND C.EMAIL LIKE '%A')  
ORDER BY C.EMAIL
```



**Q.2. LET'S INVITE THE ARTISTS WHO
HAVE WRITTEN THE MOST ROCK MUSIC
IN OUR DATASET.
WRITE A QUERY THAT RETURNS THE
ARTIST NAME AND TOTAL TRACK COUNT
OF THE TOP 10 ROCK BANDS**

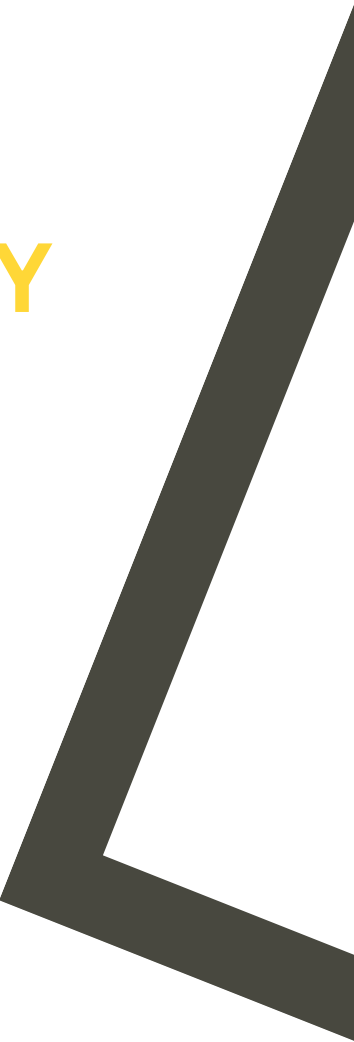
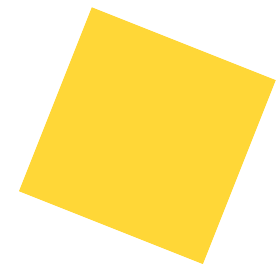
```
SELECT A.ARTIST_ID ,A.NAME,  
COUNT(A.ARTIST_ID) AS NUM_OF_SONGS  
FROM ARTIST A  
JOIN ALBUM AL ON A.ARTIST_ID = AL.ARTIST_ID  
JOIN TRACK T ON AL.ALBUM_ID = T.ALBUM_ID  
JOIN GENRE G ON T.GENRE_ID = G.GENRE_ID  
WHERE G.NAME = 'ROCK'  
GROUP BY A.ARTIST_ID  
ORDER BY NUM_OF_SONGS DESC  
LIMIT 10
```





Q3. RETURN ALL THE TRACK NAMES THAT HAVE A SONG LENGTH LONGER THAN THE AVERAGE SONG LENGTH. RETURN THE NAME AND MILLISECONDS FOR EACH TRACK. ORDER BY THE SONG LENGTH WITH THE LONGEST SONGS LISTED FIRST.

```
SELECT NAME, MILLISECONDS AS  
TIME_LENGTH_OF_SONGS  
FROM TRACK  
WHERE MILLISECONDS > (SELECT AVG(MILLISECONDS)  
FROM TRACK)  
ORDER BY MILLISECONDS DESC
```



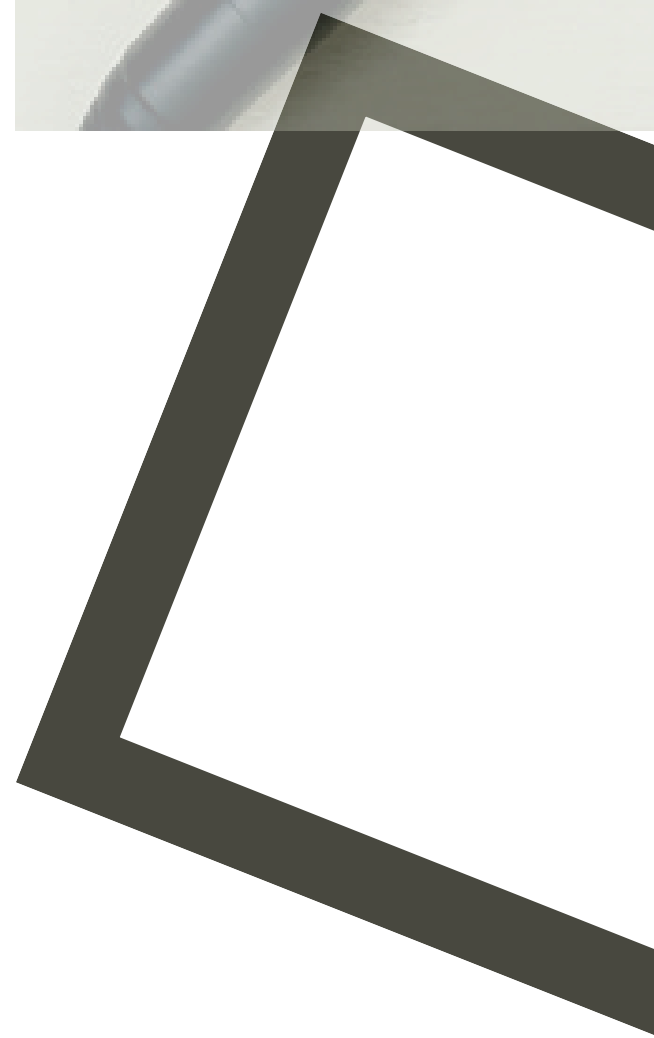
QUESTION SET - 3

ADVANCE



1. FIND HOW MUCH AMOUNT SPENT BY EACH CUSTOMER ON ARTISTS? WRITE A QUERY TO RETURN CUSTOMER NAME, ARTIST NAME AND TOTAL SPENT.

```
WITH BEST_SELLING_ARTIST AS (  
  SELECT ARTIST.ARTIST_ID AS ARTIST_ID, ARTIST.NAME AS ARTIST_NAME,  
  SUM(INVOICE_LINE.UNIT_PRICE * INVOICE_LINE.QUANTITY) AS TOTAL_SPENT  
  FROM INVOICE_LINE  
  JOIN TRACK ON TRACK.TRACK_ID = INVOICE_LINE.TRACK_ID  
  JOIN ALBUM ON ALBUM.ALBUM_ID = TRACK.ALBUM_ID  
  JOIN ARTIST ON ARTIST.ARTIST_ID = ALBUM.ARTIST_ID  
  GROUP BY 1  
  ORDER BY 3 DESC  
  LIMIT 1  
)  
SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, BSA.ARTIST_NAME,  
SUM(IL.UNIT_PRICE * IL.QUANTITY) AS TOTAL_SPENT  
FROM INVOICE INV  
JOIN CUSTOMER C ON C.CUSTOMER_ID = INV.CUSTOMER_ID  
JOIN INVOICE_LINE IL ON IL.INVOICE_ID = INV.INVOICE_ID  
JOIN TRACK T ON T.TRACK_ID = IL.TRACK_ID  
JOIN ALBUM ALB ON ALB.ALBUM_ID = T.ALBUM_ID  
JOIN BEST_SELLING_ARTIST BSA ON BSA.ARTIST_ID = ALB.ARTIST_ID  
GROUP BY 1, 2, 3, 4  
ORDER BY 5 DESC
```



Q. 2. WE WANT TO FIND OUT THE MOST POPULAR MUSIC GENRE FOR EACH COUNTRY. WE DETERMINE THE MOST POPULAR GENRE AS THE GENRE WITH THE HIGHEST AMOUNT OF PURCHASES.

WRITE A QUERY THAT RETURNS EACH COUNTRY ALONG WITH THE TOP GENRE.

FOR COUNTRIES WHERE THE MAXIMUM NUMBER OF PURCHASES IS SHARED RETURN ALL GENRES

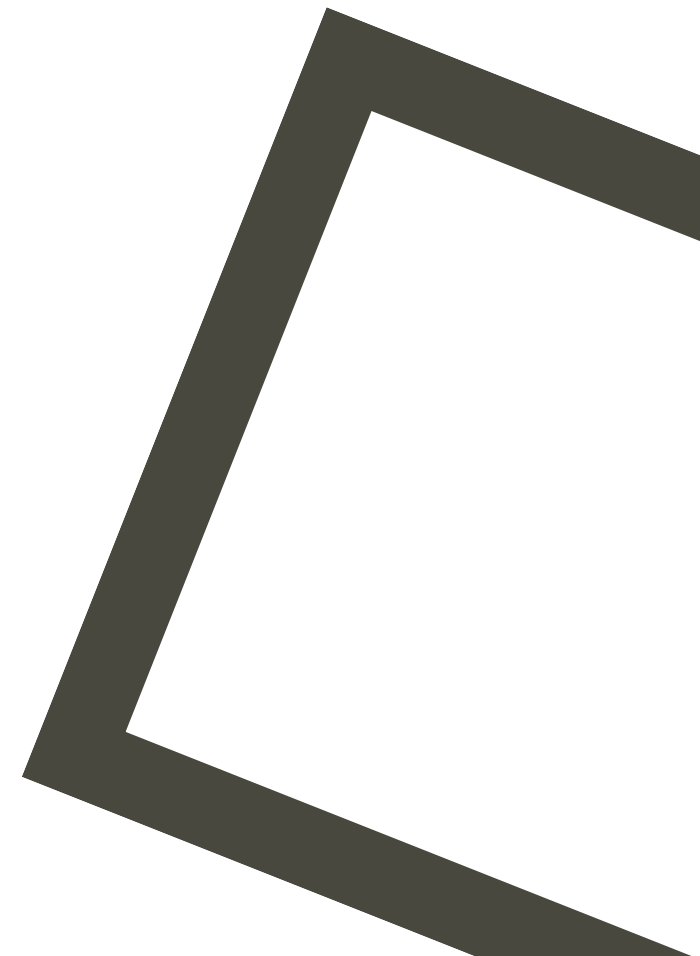
```
WITH POPULAR_GENRE AS(
SELECT
COUNT(INVOICE_LINE.QUANTITY) AS PURCHASE,
CUSTOMER.COUNTRY,
GENRE.NAME,
GENRE.GENRE_ID
,ROW_NUMBER() OVER(PARTITION BY CUSTOMER.COUNTRY ORDER BY
COUNT(INVOICE_LINE.QUANTITY) DESC ) AS ROW_NUM
FROM INVOICE_LINE
JOIN INVOICE ON INVOICE.INVOICE_ID = INVOICE_LINE.INVOICE_ID
JOIN CUSTOMER ON CUSTOMER.CUSTOMER_ID = INVOICE.CUSTOMER_ID
JOIN TRACK ON TRACK.TRACK_ID = INVOICE_LINE.TRACK_ID
JOIN GENRE ON GENRE.GENRE_ID = TRACK.GENRE_ID
GROUP BY 2, 3, 4
ORDER BY 2 ASC, 1 DESC
)
SELECT *
FROM POPULAR_GENRE
WHERE ROW_NUM <= 1
```



Q.3. WRITE A QUERY THAT DETERMINES THE CUSTOMER THAT HAS SPENT THE MOST ON MUSIC FOR EACH COUNTRY.

**-- WRITE A QUERY THAT RETURNS THE COUNTRY ALONG WITH THE TOP CUSTOMER AND HOW MUCH THEY SPENT.
-- FOR COUNTRIES WHERE THE TOP AMOUNT SPENT IS SHARED, PROVIDE ALL CUSTOMERS WHO SPENT THIS AMOUNT**

```
SELECT  
CUSTOMER.CUSTOMER_ID,  
CUSTOMER.FIRST_NAME,  
CUSTOMER.LAST_NAME,  
CUSTOMER.COUNTRY,  
SUM(INVOICE_LINE.UNIT_PRICE * INVOICE_LINE.QUANTITY) AS  
AMOUNT_SPENT  
FROM INVOICE_LINE  
JOIN INVOICE ON INVOICE.INVOICE_ID =  
INVOICE_LINE.INVOICE_ID  
JOIN CUSTOMER ON CUSTOMER.CUSTOMER_ID =  
INVOICE.CUSTOMER_ID  
GROUP BY 1, 2, 3, 4  
ORDER BY 5 DESC
```



The background is a solid yellow color. It features several abstract geometric shapes: a large white L-shaped line in the top right, a smaller dark grey square in the top right, a white L-shaped line in the bottom left, and a dark grey square in the bottom left. The main text is centered and reads "THANK YOU FOR ATTENTION".

THANK YOU FOR ATTENTION

Behind Every Dataset, There Is A Decision Waiting To Be Discovered.

WASIM RAJA